AI-Driven Healthcare Optimization: Integrating Fog and Edge Computing for Secure and Real-Time Medical Solutions

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Abstract

The advancement of new technology has compelled modern healthcare organizations through the intelligent computing modeling to enhance efficiency in processing healthcare delivery outcomes and health care organization efficiency. This chapter focuses on the application of Fog and Edge Computing integrated with Machine Learning (ML) to solve major healthcare optimization problems. It explores the decentralized computing paradigms which support real-time analytics for various use cases such as remote patient monitoring and emergency response, which leads to lower latency and increased response times. To reduce data privacy and cybersecurity threats, measures, including encryption and an ML-based anomaly detection system, are proposed. The chapter also expounds on new and creative approaches in achieving better compatibility across various healthcare systems through the approach of applying data harmonization technique built on ML and establishes tactics for developing scalable structures based on dynamic resource management in fog and edge settings. The development of adaptive ML models for providing individualized healthcare services including accurate diagnosing and treatment planning is also investigated. Furthermore, this chapter provides a detailed summary of algorithms and frameworks such as Federated Learning, Blockchain Technology, and Edge-Fog-Cloud computing that collectively permit safe and efficient real-time healthcare solutions. Each of these technologies is illustrated through ECG anomaly, medical IoT security, and patient risk prediction use cases. The chapter also considers the modern developments in artificial intelligence approaches to data analysis in the form of Data Mining, Modeling and Optimization techniques in personalized medicine, and resources management. The value of this integrative approach is to transform the current healthcare model to secure, efficient and personalized one through novel computational models.

Keywords

Healthcare · Optimization problems · Fog computing · Edge computing · Machine Learning

1 Introduction

Technology plays a critical role in societal life; it has impacted numerous fields and has particularly impacted health systems. The elements of the digital revolution have claimed to advance decision-making, diagnostics, and treatment planning as well as patient management (Aminabee 2024). Paper-based health records presented some problems in health care delivery including access and transfer of information; copying and sharing of information; and precision and accuracy of information during critical periods. A key advancement came from adopting EHR's so patients' demographic, diagnosis, and treatment information could be filed electronically in a format retrievable by the appropriate users (Nowrozy et al. 2024). However, the achievements realized when

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implementing EHRs created new barriers including high deployment costs, data privacy, and limited interoperability that challenge effective use in health care delivery (Khashan et al. 2024). To solve these problems the proposed solution is based on the integration of Fog and Edge computing with ML. Fog and edge computing implies the localisation of computational strength close to where data is being generated hence real-time data analysis is facilitated. This reduces latency and increases response time in circumstances that are critical in healthcare such as distant patient monitoring and emergency. ML algorithms strengthen this system through adding abilities of exact and accurate forecasting. Intelligent resource management, and real-time unusual activity detection for healthcare service improvement and data protection.

Fog and edge computing frameworks are advantageous for dealing with the large-scale data produced by wearables, IoMT, and mHealth. These distributed systems reduce the amount of data exchanged with the central cloud servers thus reducing the network load and increasing security. Moreover, the implemented ML models of the edge also help in delivering unique healthcare experiences by processing the patient data instantly, thus accelerating diagnosis along with providing treatment. The desire of the European Commission to have standard EHR systems is an acknowledgment of the efforts being made across the world to have interoperability of data and/or secure cross-border data sharing. Nevertheless, challenges such as either system heterogeneity, cybersecurity threats or staff resistance to adoption and use of technology remain pervasive. Still, the costs of implementing this system are high, especially in the developing countries, making widespread difficult. Moreover, the issues of data protection and patient's confidentiality are the drivers for the development of secure models within EHR systems (Tertulino et al. 2024). The interoperability concerns can also be addressed with the help of the ML techniques. In this case, the coalescing of data formats and enhancing data exchange between dissimilar healthcare systems helps guarantee that healthcare workers can obtain integrated patient data through the application of ML models. In addition, due to the opportunities offered by adaptive learning systems, staff training can be made easier and there will be less resistance to implementing new technologies. Patient identification is even now a major problem in the overall field of the health care section whereby patients are at times given wrong diagnoses. The use of such ML based identity verification systems can reduce these risks by matching the patient data between all the platforms. Moreover, to enhance CDS, striving for fewer alerts overall but those that have greater significance since they are sometimes overlooked owing to the large number created. There are ways to overcome it such as context-aware and prioritized or ML-driven alert systems that will ensure he or she does not miss essential notifications at the right time. In addition, I have identified that the combination of various kinds of wearable devices requires concerns on the data coherence and normalization. FFT and edge computing frameworks can independently or collectively collate and analyze data from different devices whereas the ML algorithms can correlate this data for better patient care. The combination of such approaches is critical for the further development of applications such as remote patient monitoring, early diagnostics, and individualized therapeutic plans. In summary, the integration of fog and edge computing with ML as a way of solving existing healthcare optimization problems is perhaps the most viable approach. This makes the health care delivery system efficient and safe resulting in improved personalized patient care to improve system strength and resilience for the future.

1.1 Contributions of the Chapter

This chapter provides substantial contributions in resolving the requirements of optimized healthcare based on fog and edge computing jointly with ML.

- Discusses the fog and edge computing models used to reduce response time and have faster results in emergency health care conditions like telemedicine and others.
- Discusses encryption, and federated learning and blockchain as top security measures to enhance the securities of medical data.
- Both stress on individual approaches of patient treatment and better diagnostic tools by ML algorithms.
- Studies various optimization algorithms for designing better flow patterns of work and utilization of resources along with minimum possible cost for healthcare systems.

2 Overview of Health Optimization

Health optimization in current and future healthcare systems involves enhancement of the delivery of services, efficiency, and successful outcome through technology. Despite the significant potential benefits of Electronic Health Records (EHRs), their deployment is associated with various costs, including those connected to, inter alia, equipment installation, upgrade, and personnel education. Another advantage of fog and edge computing is the cost-optimization because it lets process collect information with less utilization of centralized systems. Security and confidentiality are still an issue of importance because the information technology system is always prone to cybercrime. Encryption processes, physical controls, and most importantly, ML algorithms

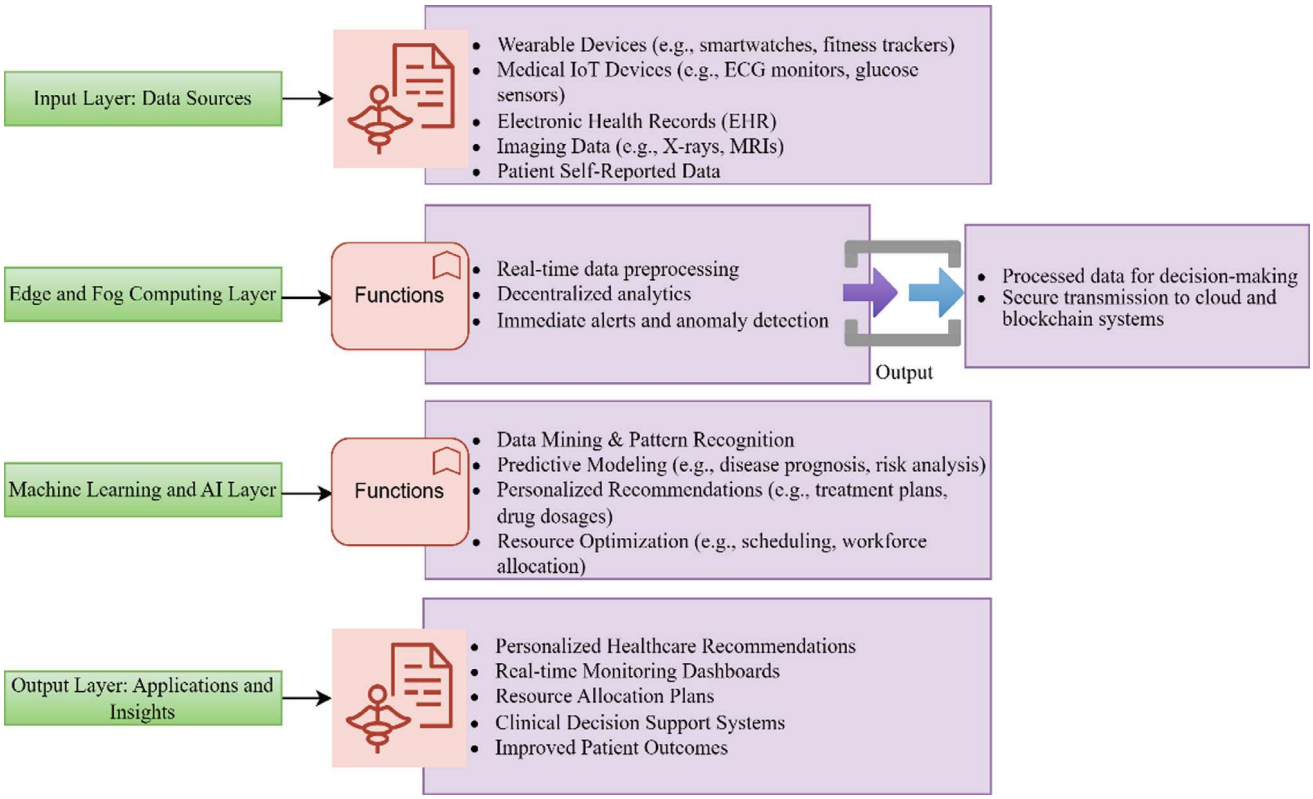


Fig. 1 AI-driven healthcare optimization framework

increase protection features. Data integration and harmonization are another problem because healthcare systems are highly heterogeneous; it can also be solved with the help of using ML algorithms for data harmonization and standardization of communication.

Figure 1 depicts a framework for healthcare optimization using edge and fog computing, ML, and AI. This is followed by an input layer which brings in data from wearable devices, medical IoT devices, EHRs, imaging data, and patient generated data. This data is then pre-processed and analyzed in the edge and fog computing layer to support real-time data preprocessing, decentralized analytics and provide immediate alerts of any anomaly. The next layer of the ML and AI for application carries out computations such as data mining, pattern analysis, model predictions, and resource allocation. The final layer is the output layer containing the presented applications and findings such as customized health care-plan recommendations, tracking interfaces, decision support tools and, at last, enhanced quality of life among patients.

The main sources of implementing new systems are staff related: on one hand, staff often resist new systems due to their comprehensiveness; on the other hand, training programs are often (resource) intensive. These problems can be solved by the usage of adaptive ML-based training modules and develop user-friendly interfaces. Patient Registry issues such as wrong identification, which might be due to keying errors, could be eliminated with an ML-powered matching of identity and record cross-check. The problem of signal fatigue in clinical decision support systems over time due to sheer number of often unactionable alerts, which reaches the clinician, can be tackled by using prioritized and contextually relevant alerts that are generated through ML algorithms. Further, the inclusion of dissimilar wearable devices into a health facility poses challenges in terms of data format especially when it must interface with different integrated systems where edge-based aggregation and ML integration can make it simple. All these strategies combine to improve the quality of health care service delivery because the core optimization factors are well tackled.

Table 1 shows the major optimization problems in the healthcare context include the following: these are high costs of EHR implementation; issues on data privacy and security; system incompatibility; there could be resistance to change from the staff; identification of the patients; clinician alerts fatigue in CDS; and, lastly, integration of the many wearables. These solutions require fog and edge computing solutions alongside with ML technologies like distributed processing for less cost, data encryption, data formatting, self-tuning

Table 1 Major barriers and approaches to healthcare optimization of Fog and Edge computing in Machine Learning

Optimization problem	Cause	Proposed solution (Using Fog/Edge and ML)	References
High Implementation Costs	Significant expenses for setup, maintenance, and staff training	Cost-effective fog/edge solutions for distributed data processing, reducing infrastructure needs	Khashan et al. (2024)
Data Privacy and Security Risks	Vulnerability to cyberattacks and data breaches	Implement robust encryption, access controls, and ML-based anomaly detection for enhanced security	Tertulino et al. (2024)
Interoperability & Standardization Issues	Heterogeneous systems hindering data sharing and integration	ML-driven data harmonization and standardized communication protocols for seamless data exchange	Begoyan (2007)
Staff Resistance & Training Challenges	Complex systems and costly, time-intensive training	User-friendly interfaces and adaptive ML-based training modules for improved user experience and reduced training burden	Downing et al. (2018)
Patient Misidentification	Errors in patient registration and data entry	ML-based identity verification and cross-referencing patient records for accurate identification and data integrity	Seiden and Barach (2006)
Alert Fatigue in Clinical Decision Support	Overloaded alerts leading to ignored notifications	Prioritized and context-aware ML-driven alert systems to reduce alert overload and improve clinician focus	Carspecken et al. (2013)
Integration of Wearable Devices	Diverse devices and data formats causing inconsistency	Edge-based aggregation and ML integration for seamless data processing and analysis from various wearable devices	Sempionatto et al. (2020)

training blocks, IDMS, alert triage and incorporating wearable device data into the healthcare system for efficiency and security.

3 Machine Learning Techniques for Health Optimization

ML is quickly becoming an important tool in healthcare due to the ability to diagnose an individual, tailor a treatment plan for them, and even manage resources well. XGBoost works best for prognosticating patient health while for disease categorization and patient data classification, SVM is the best algorithm. Methods implemented in today's world such as Random Forest and PCA increase the efficiency and readability of studies. JADE, Enhanced AEO, LevyJA, And the Original Jaya Algorithm are among the optimization algorithms that solve advanced healthcare problems including resource allocation and treatment planning. All these tools mentioned above can help healthcare providers enhance patient satisfaction, decrease expenditure, and enhance research.

Table 2 outlines several ML techniques and their applications in health optimization. Techniques like XGBoost excel at making accurate predictions for patient health monitoring. SVM effectively classifies diseases and patient data. Random Forest improves prediction accuracy by combining multiple decision trees. PCA simplifies complex health data by reducing its dimensionality. Optimization algorithms like JADE, Enhanced AEO, LevyJA, and the Original Jaya Algorithm efficiently address various healthcare challenges, including resource allocation, treatment planning, and improving the accuracy of optimization tasks. These techniques collectively contribute to advancements in healthcare by enabling more accurate diagnoses, personalized treatments, and efficient resource management.

Table 3 presents an overview of the optimization using AI in the health sector. Business intelligence technique such as data mining and pattern recognition help in discovery of patterns within patient data in helping in disease diagnosis and assessment of treatment effectiveness. Outcome predictions of diseases were made to facilitate early and planned control measures and risk estimating. AI used in the development of personalized medicine ensures that the treatment being offered meets the patient's particular needs thus enhancing the treatments efficiency. Last but not the least, optimization algorithms tying up resources in the healthcare sector provide increased efficiency, lower wait times. Altogether, these techniques have their positive effects on the patients and healthcare, as well as on the rational use of resources.

Table 2 Machine Learning techniques for health optimization

Machine Learning technique	Description	Application in health optimization	References
Extreme Gradient Boosting (XGBoost)	A highly accurate method for making predictions by combining and improving upon a previous method called Gradient Boosting	Predictive analytics for patient health monitoring and disease progression	Shehab et al. (2024)
Support Vector Machine (SVM)	A model that effectively classifies data by using hyperplanes and a technique called "kernel trick" to handle both simple and complex data patterns	Disease classification, diagnostic support, and patient data categorization	Reddy Madhavi et al. (2023), Maharani and Atastina (2021)
Random Forest	A method that combines multiple decision trees to make more accurate predictions by using a majority vote system	Risk prediction, patient outcome forecasting, and anomaly detection in medical data	James et al. (2013), Harris and Grunsky (2015), Khajavi and Rastgoo (2023)
Principal Component Analysis (PCA)	A technique that simplifies complex health data by reducing the number of variables while preserving the most important information	Simplifying complex health data for better visualization and analysis	Lid and Planning (1993), Kurita (2021), Kherif and Latypova (2019)
JADE (Adaptive Differential Evolution)	An adaptive algorithm that dynamically adjusts its parameters to efficiently solve optimization problems in healthcare, such as resource allocation	Optimal resource allocation in health systems and improving the efficiency of diagnostic algorithms	Zhang et al. (2007), Zhang and Sanderson (2009)
Enhanced Adaptive Epsilon Optimization (Enhanced AEO)	An improved version of AEO that effectively balances exploration and exploitation to optimize medical data analysis and decision-making	Efficient medical data analysis and optimized decision-making	Thieu and Mirjalili (2023), Eid et al. (2020)
Levy Flight Jaya Algorithm (LevyJA)	An optimization algorithm that incorporates a special search strategy (Levy flight) to effectively solve complex healthcare problems, such as treatment planning	Solving complex healthcare problems, such as resource scheduling and treatment planning	Rao (2016), Iacca et al. (2021), Sathish Kumar et al. (2023)
Original Jaya Algorithm (OriginalJA)	A simple yet effective optimization algorithm that can find the best solutions for a wide range of healthcare problems without requiring any manual parameter tuning	Improving accuracy and efficiency in healthcare optimization tasks	Ke et al. (2023), Kumar and Jain (2024)

Table 3 AI-driven optimization techniques in healthcare

AI-driven technique	Purpose and functionality	Healthcare impact	References
Data Mining and Pattern Recognition	Uncovers hidden patterns within large datasets using techniques like association rule mining, clustering, and classification	Identifies disease prevalence, assesses treatment efficacy, and improves clinical decision-making processes	Kuruba et al. (2023)
Predictive Analytics for Diagnosis	Employs methods like logistic regression, decision trees, and ensemble models to forecast disease outcomes	Enables early disease detection, identifies risk factors, and allows for patient stratification to facilitate timely interventions	Badawy et al. (2023)
Personalized Medicine and Treatment Systems	Develops AI-driven recommendations for personalized treatment plans based on individual patient data	Tailors treatments to individual needs, enhancing treatment effectiveness and improving patient adherence to treatment plans	Çelik Ertuğrul and Elçi (2020)
Optimization Algorithms for Resource Management	Utilizes techniques such as linear programming, integer programming, and simulation modeling to optimize resource allocation	Streamlines resource allocation within healthcare systems, reduces patient wait times, and enhances overall operational efficiency	Wang and Demeulemeester (2023)

3.1 Algorithms and Frameworks for Health Optimization

The Algorithms and Frameworks for Health Optimization section showcases how metaheuristic algorithms can solve some of the complications surrounding the improvement of healthcare. Some soft computing techniques like Genetic Algorithms (GA), Particle Swarm Optimization (PSO), and Ant Colony Optimization (ACO) work well in managing resources, they schedule tasks and optimize the network routes in the healthcare system. Algorithms like CS, GWO, and WOA improve the predictive healthcare models, feature selection in data, and organize patient data effectively. In addition, there are other metaheuristic algorithms, for

Table 4 Algorithms and frameworks for health optimization

Algorithms	Inspiration/concept	Application in health optimization	References
GA	Natural selection and evolutionary principles	Scheduling, optimizing resource allocation in distributed healthcare systems	Katoch et al. (2021)
PSO	Social behavior of birds and fish (flocking, schooling)	Resource allocation, identifying important features, and optimizing solutions in healthcare	Shi and Eberhart (1999)
ACO	Foraging behavior of ants	Optimizing network routing, pathfinding, and task scheduling in medical systems	Dorigo et al. (2006)
CS	Breeding behavior of cuckoos (brood parasitism)	Solving complex healthcare problems and optimizing solutions in engineering-based healthcare applications	Gandomi et al. (2013)
GWO	Social hierarchy and hunting strategies of grey wolves	Selecting important features and optimizing parameters in medical diagnostics	Mirjalili et al. (2014)
WOA	Bubble-net hunting technique of humpback whales	Predictive healthcare modeling and grouping similar patient data (clustering)	Mirjalili et al. (2014), Gharehchopogh and Gholizadeh (2019)
BA	Foraging behavior of honeybees	Optimizing medical logistics and the distribution of healthcare resources	Karaboga and Akay (2009)
HHO	Cooperative hunting strategies of Harris hawks	Optimizing scheduling and decision-making processes within healthcare systems	Alabool et al. (2021), Heidari et al. (2019)
MFO	Moths' behavior of flying towards light sources	Solving complex medical problems and selecting important features in disease prediction	Mirjalili (2015)

example: the Bee Algorithm (BA), Harris Hawks Optimization (HHO), and Moth-Flame Optimization (MFO), which increases health care service delivery, helps to avoid errors in decision-making processes, and improves the accuracy of disease diagnosing. Altogether, these methods help in the attainment of efficient healthcare since real-time analysis of data, as well as dynamic adjustment of decision-making are made possible, and overall system performance is enhanced especially under fog and edge computing settings.

Table 4 provides a cross-section of algorithms and frameworks from nature working toward health optimization. GA are based on the natural selection, while PSO belongs to the group based on the bird and fish flock. That is ACO simulates the foraging behavior of ants while CS simulates the brood parasitism of cuckoos. GWO mimics the grey wolf used in their hunting strategies and WOA was derived from the humpback whales' bubble-nets. The BA reflects the honeybee foraging process, and HHO reflects the cooperative hunting process of the Harris hawks. Last, MFO is inspired from living nature more specifically from moth flying toward the flame. Some of its uses include resource management, feature extraction, scheduling and solving medical issues.

Table 5 explains several technologies and frameworks that are transforming healthcare. FL allows the training of the global model with several participating clients' data without the exchange of the data itself. Blockchain technology improves the security and the levels of information sharing. Edge-Fog-Cloud models help to process data and minimize the time gap. Here, Block Learning Framework combines the use of blockchain with FL to develop secure data sharing. Generating fiduciary responsibility of shared data, FLChain motivates and monitors data contributions in a federated learning environment. The Differential Privacy techniques to ensure that individual's privacy rights are not infringed during the analysis of data. Some of the properties of consensus algorithms include Consensus algorithms make all nodes agree. Homomorphic Encryption allows working with encrypted data, while preserving the privacy of the customer. Resource Trading Systems can be used to incentivise edge computing networks participation. Finally, Smart Contracts control and enable data retrieval and exchange in medical IoT systems. Each one of these technologies collectively provides secure, efficient, and privacy-preserving solutions in the healthcare sector.

4 Case Studies in Fog and Edge Computing in Healthcare

In the following part of the present chapter, it is possible to observe several examples from the reality which show that fog and edge computing can be implemented to improve the given form of healthcare services. One such case is based on the deployment of edge computing for real-time analysis of Electrocardiogram (ECG). This is due to a decentralized data processing technique known as federated learning (FL), which ensures privacy and latency problems of centralzed

Table 5 Key technologies and frameworks for secure and efficient healthcare systems

Technology/framework	Concept/functionality	Healthcare application	References
Federated Learning (FL)	Training ML models without centralizing data	Detecting anomalies in ECG signals, privacy-preserving analysis of medical data	Jin and Dong (2018)
Blockchain Technology	An immutable and decentralized ledger for secure data storage and sharing	Enhancing security and transparency in medical IoT systems, particularly for federated learning	Cheikhrouhou et al. (2021)
Edge-Fog-Cloud Computing	A layered architecture for processing data closer to the source (edge devices) to reduce latency	Real-time patient monitoring and efficient processing of medical data	Zhang et al. (2020)
BlockLearning Framework	A modular framework combining blockchain with federated learning	Securely sharing data across different organizations while maintaining privacy (vertical federated learning)	Dias and Meratnia (2022)
FLChain Ecosystem	A decentralized and incentivized system for federated learning	Enabling publicly auditable and secure sharing of medical data	Bao et al. (2019)
Differential Privacy Techniques	Methods for adding noise to data to protect individual privacy	Securing model updates in federated learning by preventing the leakage of sensitive patient information	Aich et al. (2021)
Consensus Algorithms	Algorithms that enable distributed agreement among multiple parties	Coordinating model updates in healthcare applications within a decentralized environment	Lu et al. (2020)
Homomorphic Encryption	Allows for computations to be performed on encrypted data without the need to decrypt it	Enabling privacy-preserving aggregation of data in medical federated learning	Chen et al. (2020)
Resource Trading Systems (Hybrid)	Systems that incentivize participation in edge computing networks through blockchain-based rewards	Encouraging the sharing of healthcare data among diverse edge nodes	Fan et al. (2020)
Smart Contracts	Self-executing contracts on blockchain that automate the execution of predefined conditions	Automating data access and sharing within medical IoT systems	Ali et al. (2021)

systems are avoided. This allows early identification of cardiovascular irregularities while ensuring that highly determinable patient information goes only through local machines (Prabhu and Hanumanthaiah 2022). Another massive use-case relates to the combination of blockchain technology with FL in medical IoT appliances. In their framework proposed in Cherukuri et al. (2024), they introduce blockchain for the purposes of strengthening security and promoting the transparency of model training in collaborative settings. Such an approach avoids leaking data and offers a safe and distributed way to compare numerous model updates—a major privacy issue in medicine. In addition, (Dias and Meratnia 2022) presented the BlockLearning framework that uses federated vertical learning along with the blockchain technique to maintain and analyze the patient data of multiple hospitals securely. It makes secure interaction between various organizations or even departments of a large healthcare network possible, showing that such systems can maintains data confidentiality, too. However, a study in Zhang et al. (2020) presented an edge-fog-cloud computing model for real-time patient monitoring. The three-tier design of the SmartPG also load balances the computational load in the edge and fog nodes to reduce latency while storing big data in the cloud. It enhances response time in emergency situations and generates the maximum utilization of available health resources. Taken together, these case studies illustrate that the combination of fog and edge computing with other forms of innovative technologies such as blockchain and federated learning has a vast capacity. These integrates result in better healthcare service delivery, increase in data privacy and effective health monitoring.

These studies are summarized in Table 6 in terms of approach, strength, and weakness of HLHCDA. It includes secondary uses such as epidemiological analysis on patients based on EHRs, self-reported firearm violence, and using big data and ML for readmission prediction of patients. Other works are dedicated to the use of XGBoost for risk assessment, artificial intelligence in chronic heart failure, and prescriptive analytics of medications. Further, the table examines EHR Interoperability solutions, Resource allocation optimization algorithms, Deep Medical Imaging using DL, and Predictive Analytics of patient risk stratification. Despite the insight that this research provides into the potential of applying data analytics to healthcare, there are problems such as data constraints, computational expense, and data quality requirements.

Table 6 Key studies in healthcare data analytics: approaches, advantages, and limitations

Approach	Advantages	Limitations	References
Organizing and visualizing electronic health record (EHR) data	Provides a better understanding of patient groups and disease patterns	Can only analyze past data	Hohman et al. (2023)
Evaluating self-reported firearm violence	Assesses characteristics related to demographics and clinical factors	Limited to specific patient-reported information	Cook et al. (2024)
Using ML to predict hospital readmissions	Detects patients at risk of readmission early on; reduces readmission rates	Requires a large amount of historical data	Lo et al. (2021)
Using a specific ML model (XGBoost) to predict readmissions	Effectively identifies high-risk pediatric patients for readmission	Limited to analyzing characteristics of pediatric patients	Silva et al. (2024)
Using AI for managing heart failure	Improves healthcare processes and personalizes treatment strategies	Results from initial studies may vary	Sideris et al. (2024)
Using advanced analytics to manage medications	Provides personalized recommendations for medication use	Heavily depends on the accuracy of data	Bastani and Shi (2023)
Developing solutions for EHR interoperability	Improves access to data by integrating different EHR systems	Faces ongoing challenges in achieving seamless interoperability	Derecho et al. (2024)
Using algorithms to optimize resource allocation	Improves efficiency and reduces costs	May not be able to meet all resource demands	Sharif et al. (2023)
Using deep learning for analyzing medical images	Achieves high accuracy in detecting diseases	Requires significant computing power	Rana and Bhushan (2023)
Using predictive analytics to categorize patients based on risk	Enables proactive patient management and resource allocation	Relies on having access to high-quality data	Ruiz-Ramos et al. (2024)

5 Future Research Directions

The potential areas of developments for the future studies in improving healthcare through Fog and Edge Computing with ML include Optimizing the scalability and resiliency of the decentralized systems particularly the one that is accompanying the huge IoT and medical devices data. To bring such services to the edge, and to be able to execute AI models on the restrained resources of edge devices, lightweight AI models will have to be created. Other serious line of defence methods that must be researched include differential privacy and secure multi-party computation. Besides, incorporating quantum computing to federated learning could provide more computing capabilities to address different healthcare problems. Lastly, only experimental probe investigations of such systems can assess their socio-technical impacts while advocating for compliance with acceptable ethical norms and the intake capacity of various healthcare systems worldwide.

6 Chapter Summary

In combination with Fog and Edge Computing, ML as a methodology has become one of the key disruptive strategies to major healthcare issues. These technologies help to decentralize data processing and to produce real-time analytics, which decreases latency, optimizes the patient's condition control, and consequently contributes to the timely provision of medical assistance. Introducing innovative techniques like, Federated learning, Blockchain guarantees that the flow of medical data is protected from privacy and cyberattacks. Automated solutions also improve the biometric analysis, diagnostics, and offers precisely customized treatments and preventive care for illnesses. In addition, optimization of algorithms in the management of health resources leads to the optimization of resource utilization, reduction of operating expenses and overall improvement of service provision. The integration of modern computational sciences with strongly prospective and at the same time safe and sustainable technologies implies a transition to a new level of heath care. In this chapter, very beautiful benefits harvested from the conjunction of Fog and Edge Computing with AI optimization techniques in healthcare applications have been presented,

creating the possibility to open for the coming of new quite innovative ideas and solutions that aim at providing better, safe, effective, and patient-centered medical services.

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